

Lecture 4

SIGNAL SPACE ANALYSIS

Outline

- *Introduction*
- Geometric Representation of Signals
 - Gram-Schmidt Orthogonalization Procedure
- Conversion of the AWGN into a Vector Channel
- Maximum Likelihood Decoding
- Correlation Receiver
- Probability of Error

Introduction – the Model

- We consider the following model of a generic transmission system (*digital source*):
 - A message source transmits 1 symbol every T sec
 - Symbols belong to an alphabet M ($m_1, m_2, \dots, m_M$)
 - Binary – symbols are 0s and 1s
 - Quaternary PCM – symbols are 00, 01, 10, 11



Transmitter Side

- Symbol generation (message) is probabilistic, with a priori probabilities $p_1, p_2, \dots, p_M$. or
- Symbols are equally likely
- So, the probability that symbol m_i will be emitted:

$$\begin{aligned}\rho_i &= P(m_i) \\ &= \frac{1}{M} \text{ for } i=1,2,\dots,M\end{aligned}\quad (1)$$

- Transmitter takes the *symbol (data) m_i* (digital message source output) and encodes it into a *distinct signal $s_i(t)$* .
- The *signal $s_i(t)$* occupies the whole *slot T* allotted to *symbol m_i* .
- $s_i(t)$ is a real valued energy signal (???)

$$E_i = \int_0^T s_i^2(t) dt, \quad i=1,2,\dots,M \quad (2)$$

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- The *signal $s_i(t)$* occupies the whole *slot T* allotted to *symbol m_i* .
- $s_i(t)$ is a real valued energy signal (*signal with finite energy*)

$$E_i = \int_0^T s_i^2(t) dt, \quad i=1,2,\dots,M \quad (2)$$

Channel Assumptions:

- *Linear*, wide enough to accommodate the signal $s_i(t)$ with no or negligible distortion
- *Channel noise* is $w(t)$ is a zero-mean white Gaussian noise process – AWGN
 - additive noise
 - received signal may be expressed as:

$$x(t) = s_i(t) + w(t), \quad \left\{ \begin{array}{l} 0 \leq t \leq T \\ i=1,2,\dots,M \end{array} \right\} \quad (3)$$

Receiver Side

- Observes the received signal $x(t)$ for a duration of time T sec
- Makes an estimate of the transmitted signal $s_i(t)$ (eq. symbol m_i).
- Process is statistical
 - presence of noise
 - errors
- So, receiver has to be designed for *minimizing the average probability of error (P_e)*

$$P_e = \sum_{i=1}^M p_i P(\hat{m} \neq m_i / m_i) \quad (4)$$

The diagram illustrates the components of the equation for the average probability of error, P_e . It features three orange callout boxes with lines pointing to specific parts of the equation:

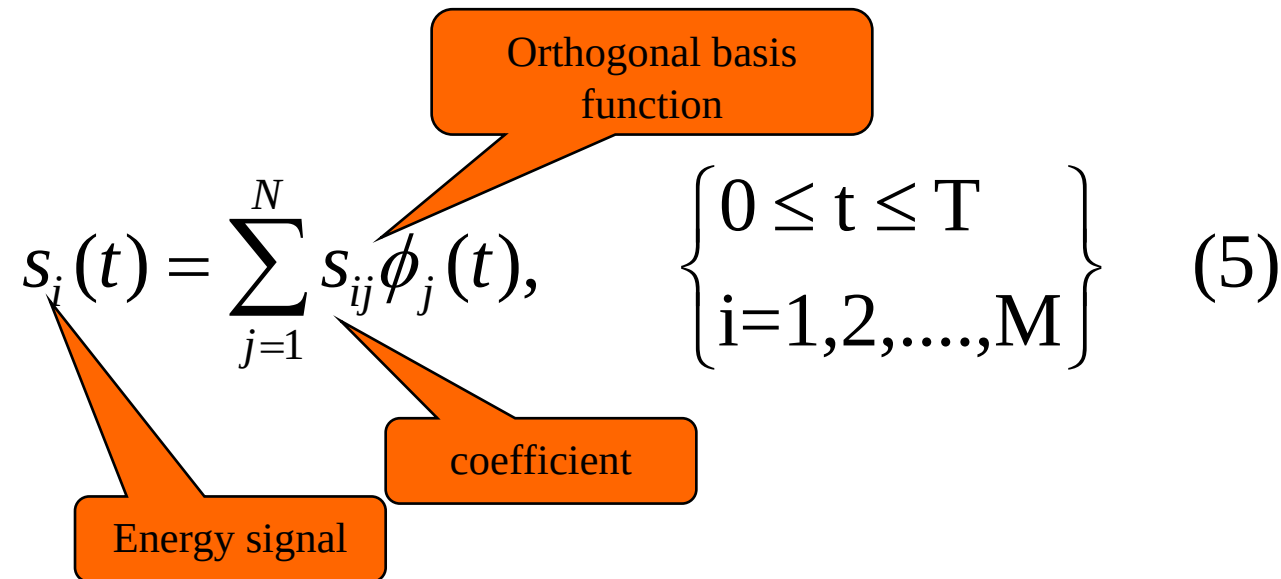
- A box labeled "What is this?" points to the summation symbol $\sum$.
- A box labeled "Symbol sent" points to the term m_i in the denominator of the conditional probability $P(\hat{m} \neq m_i / m_i)$.
- A box labeled "cond. error probability given ith symbol was sent" points to the entire conditional probability term $P(\hat{m} \neq m_i / m_i)$.

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Geometric Representation of Signals

- Objective: To represent any set of **M energy signals** $\{s_i(t)\}$ as linear combinations of **N orthogonal basis functions**, where $N \leq M$
- Real value energy signals $s_1(t), s_2(t), \dots, s_M(t)$, each of duration T sec

$$s_i(t) = \sum_{j=1}^N s_{ij} \phi_j(t), \quad \left\{ \begin{array}{l} 0 \leq t \leq T \\ i=1,2,\dots,M \end{array} \right\} \quad (5)$$


The diagram illustrates the equation $s_i(t) = \sum_{j=1}^N s_{ij} \phi_j(t)$ with three callout boxes: 'Energy signal' pointing to $s_i(t)$, 'coefficient' pointing to s_{ij} , and 'Orthogonal basis function' pointing to $\phi_j(t)$. The domain $\left\{ \begin{array}{l} 0 \leq t \leq T \\ i=1,2,\dots,M \end{array} \right\}$ is also shown.

- Coefficients:

$$s_{ij} = \int_0^T s_i(t)\phi_j(t)dt, \quad \left\{ \begin{array}{l} i=1,2,\dots,M \\ j=1,2,\dots,M \end{array} \right\} \quad (6)$$

- Real-valued basis functions:

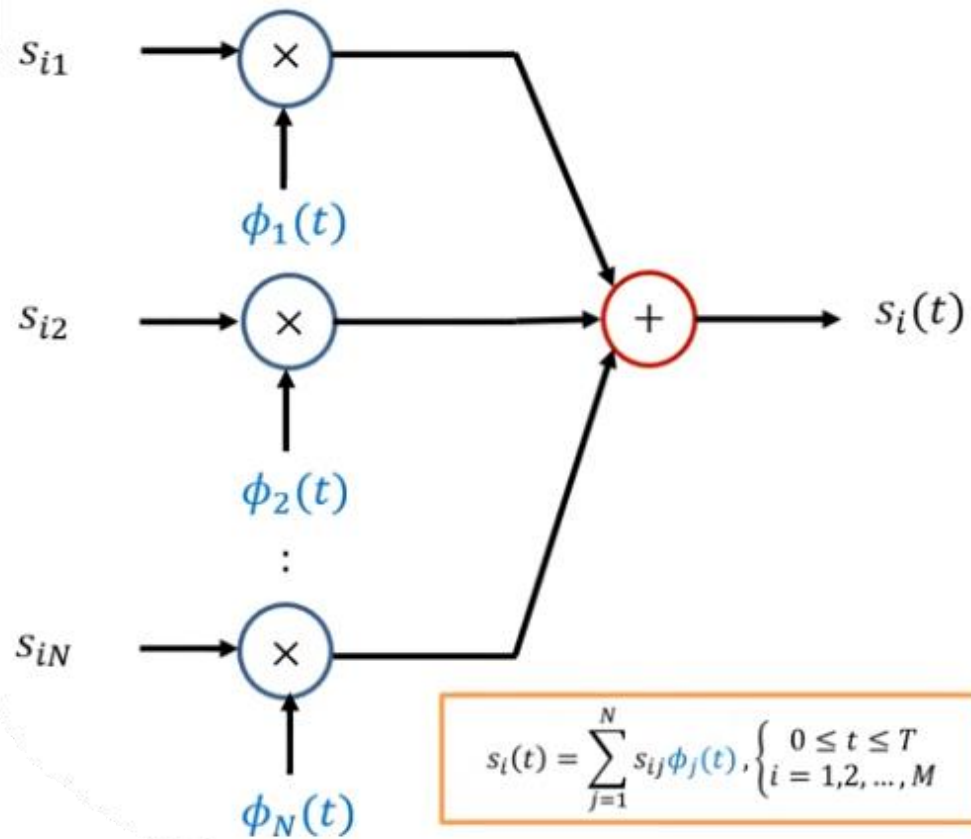
$$\int_0^T \phi_i(t)\phi_j(t)dt = \delta_{ij} = \left\{ \begin{array}{ll} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{array} \right\} \quad (7)$$

- The set of coefficients can be viewed as a N-dimensional vector, denoted by s_i
- Bears a one-to-one relationship with the transmitted signal $s_i(t)$

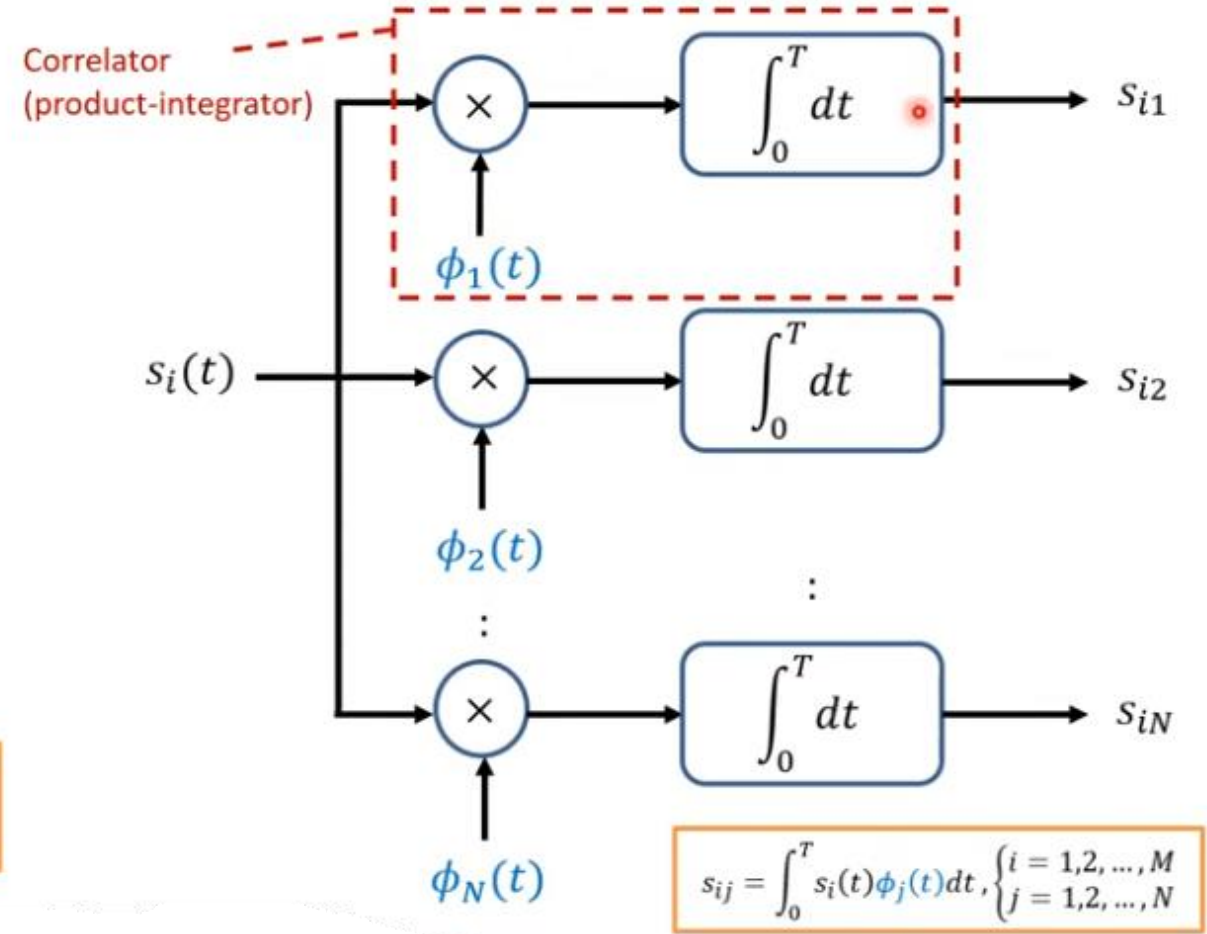
(a) Synthesizer for generating the signal $s_i(t)$.

(b) Analyzer for generating the set of signal vectors $\{s_i\}$.

Synthesizer for generating the signal $s_i(t)$



Analyzer for generating the elements of s_i



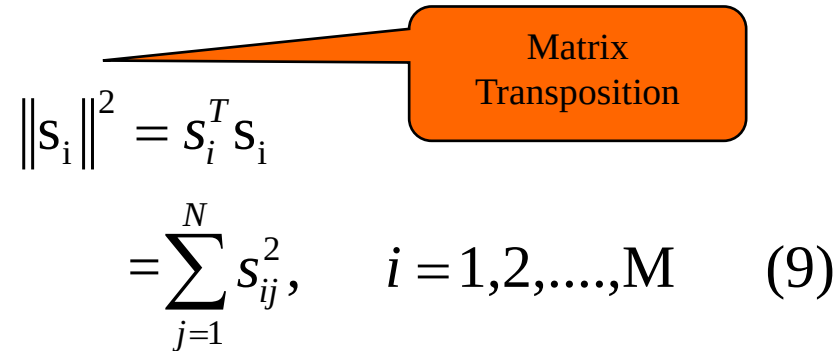
So,

- Each signal in the set $s_i(t)$ is completely determined by the vector of its coefficients

$$s_i = \begin{bmatrix} s_{i1} \\ s_{i2} \\ \cdot \\ \cdot \\ \cdot \\ s_{iN} \end{bmatrix}, \quad i = 1, 2, \dots, M \quad (8)$$

Finally,

- The signal vector $\mathbf{s}_i$ concept can be extended to 2D, 3D etc. N-dimensional Euclidian space
- Provides mathematical basis for the geometric representation of energy signals that is used in noise analysis
- Allows definition of
 - Length of vectors (absolute value)
 - Angles between vectors
 - Squared value (inner product of $\mathbf{s}_i$ with itself)

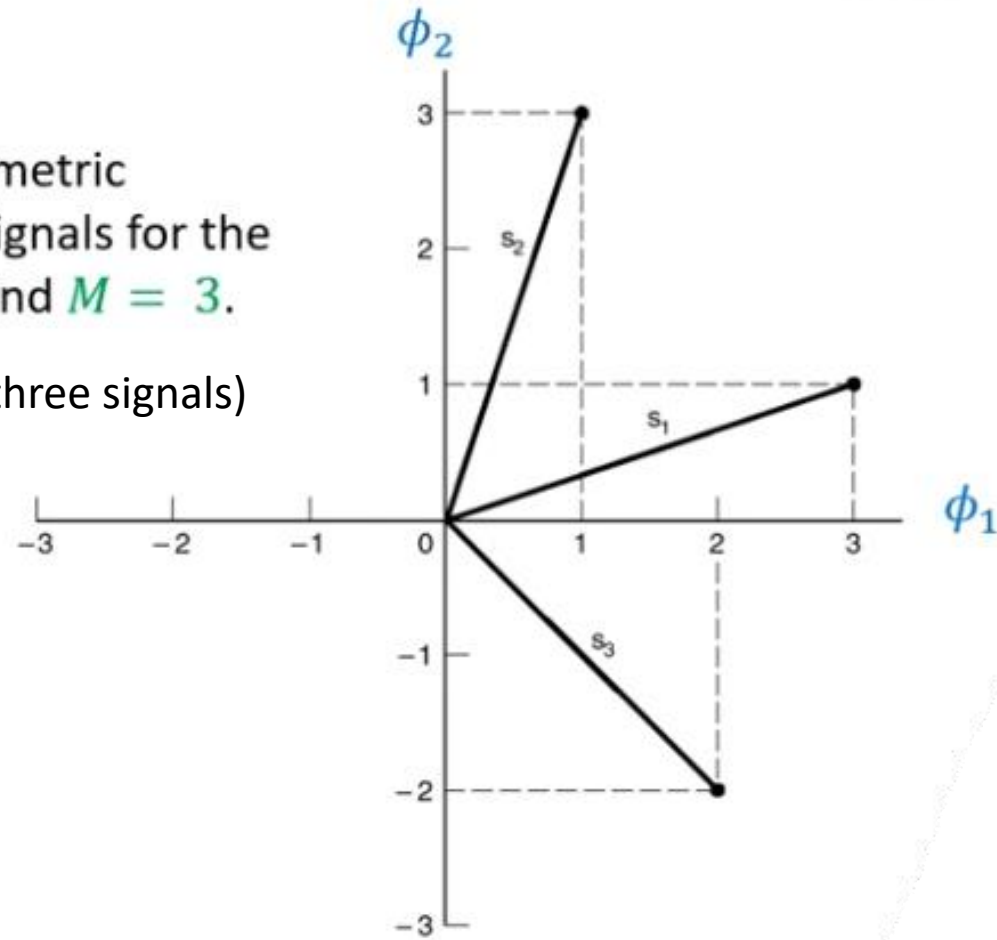


The diagram illustrates the calculation of the squared length of a vector $\mathbf{s}_i$. It features the equation $\|\mathbf{s}_i\|^2 = \mathbf{s}_i^T \mathbf{s}_i$ followed by $= \sum_{j=1}^N s_{ij}^2$, with $i = 1, 2, \dots, M$ and a label (9). An orange callout box labeled "Matrix Transposition" points to the $\mathbf{s}_i^T$ term in the first equation.

$$\|\mathbf{s}_i\|^2 = \mathbf{s}_i^T \mathbf{s}_i$$
$$= \sum_{j=1}^N s_{ij}^2, \quad i = 1, 2, \dots, M \quad (9)$$

Illustrating the geometric representation of signals for the case when $N = 2$ and $M = 3$.

(two-dimensional space, three signals)



Also,

What is the relation between the *vector representation* of a signal and its *energy value*?

- ...start with the definition of average energy in a signal...(10)

$$E_i = \int_0^T s_i^2(t) dt \quad (10)$$

- Where $s_i(t)$ is as in (5):

$$s_i(t) = \sum_{j=1}^N s_{ij} \phi_j(t), \quad (5)$$

- After substitution:
$$E_i = \int_0^T \left[\sum_{j=1}^N s_{ij} \phi_j(t) \right] \left[\sum_{k=1}^N s_{ik} \phi_k(t) \right] dt$$

- After regrouping:
$$E_i = \sum_{j=1}^N \sum_{k=1}^N s_{ij} s_{ik} \int_0^T \phi_j(t) \phi_k(t) dt \quad (11)$$

- $\Phi_j(t)$ is orthogonal, so finally we have:
$$E_i = \sum_{j=1}^N s_{ij}^2 = \|\mathbf{s}_i\|^2 \quad (12)$$

The energy of a signal is equal to the squared length of its vector

Formulas for two signals

- Assume we have a pair of signals: $s_i(t)$ and $s_j(t)$, each represented by its vector,
- Then:

$$s_{ij} = \int_0^T s_i(t)s_k(t)dt = s_i^T s_k \quad (13)$$

Inner product of the signals is equal to the inner product of their vector representations $[0,T]$

Inner product is invariant to the selection of basis functions

Euclidian Distance

- The Euclidean distance between two points represented by vectors (signal vectors) is equal to $\|s_i - s_k\|$ and the squared value is given by:

$$\begin{aligned}\|s_i - s_k\|^2 &= \sum_{j=1}^N (s_{ij} - s_{kj})^2 \\ &= \int_0^T (s_i(t) - s_k(t))^2 dt\end{aligned}\tag{14}$$

Angle between two signals

- The *cosine of the angle* θ_{ik} between two signal vectors s_i and s_k is equal to the inner product of these two vectors, divided by the product of their norms:

$$\cos \theta_{ik} = \frac{s_i^T s_k}{\|s_i\| \|s_k\|} \quad (15)$$

- So the two signal vectors are *orthogonal* if their inner product $s_i^T s_k$ is zero ($\cos \theta_{ik} = 0$)

Schwartz Inequality

- Defined as:

$$\left(\int_{-\infty}^{\infty} s_1(t)s_2(t)dt \right)^2 = \left(\int_{-\infty}^{\infty} s_1^2(t)dt \right) \left(\int_{-\infty}^{\infty} s_2^2(t)dt \right) \quad (16)$$

- accept without proof...

Exercise: Geometric Representation of Signals

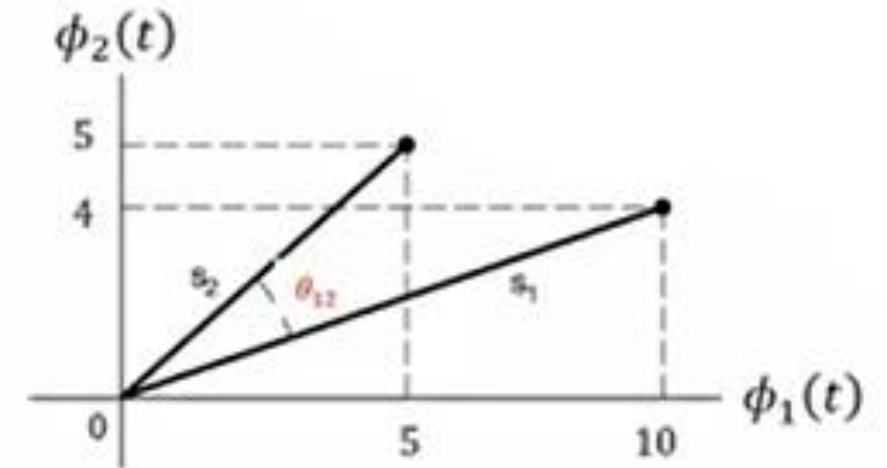
- Show that

$$\int_0^T s_i(t)s_k(t)dt = \mathbf{s}_i^T \mathbf{s}_k$$

- which is the inner product of $s_i(t)$ and $s_k(t)$ over the interval $[0, T]$
- Note that the inner product of two signals $s_x(t)$ and $s_y(t)$ does **not** depend on the choice of basis functions $\{\phi_j(t)\}$, and it only depends on the components of the signals $s_x(t)$ and $s_y(t)$ **projected** onto each of the basis functions.

Exercise

- For the geometric representation shown and assuming orthonormal basis.
 - Write S_1 and S_2 in terms of the basis functions.
 - Find the norm of S_1
 - Find the energy in the in S_2
 - Find the angle between the two vectors, S_1 and S_2 .



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Gram-Schmidt Orthogonalization Procedure

Assume a set of M energy signals denoted by $s_1(t)$, $s_2(t)$, .. , $s_M(t)$.

1. Define the first basis function starting with s_1 as: (where E is the energy of the signal)
2. Then express $s_1(t)$ using the basis function and an energy related coefficient s_{11} as:
3. Later using s_2 define the coefficient s_{21} as:

$$\phi_1(t) = \frac{s_1(t)}{\sqrt{E_1}} \quad (19)$$

$$s_1(t) = \sqrt{E_1} \phi_1(t) = s_{11} \phi_1(t) \quad (20)$$

$$s_{21} = \int_0^T s_2(t) \phi_1(t) dt \quad (21)$$

4. If we introduce the intermediate function g_2 as:

$$g_2(t) = s_2(t) - s_{21}\phi_1(t) \quad (22)$$

Orthogonal to $\phi_1(t)$

5. We can define the second basis function $\phi_2(t)$ as:

$$\phi_2(t) = \frac{g_2(t)}{\sqrt{\int_0^T g_2^2(t)dt}} \quad (23)$$

6. Which after substitution of $g_2(t)$ using $s_1(t)$ and $s_2(t)$ it becomes:

$$\phi_2(t) = \frac{s_2(t) - s_{21}\phi_1(t)}{\sqrt{E_2 - s_{21}^2}} \quad (24)$$

• Note that $\phi_1(t)$ and $\phi_2(t)$ are orthogonal that means:

$$\int_0^T \phi_2^2(t)dt = 1 \quad (\text{Look at 23})$$

$$\int_0^T \phi_1(t)\phi_2(t)dt = 0$$

And so on for N dimensional space...,

- In general a basis function can be defined using the following formula:

$$g_i(t) = s_i(t) - \sum_{j=1}^{i-1} s_{ij} \phi_j(t) \quad (25)$$

- where the coefficients can be defined using:

$$s_{ij} = \int_0^T s_i(t) \phi_j(t) dt, \quad j = 1, 2, \dots, i-1 \quad (26)$$

Special case:

- For the special case of $i = 1$ $g_i(t)$ reduces to $s_i(t)$.

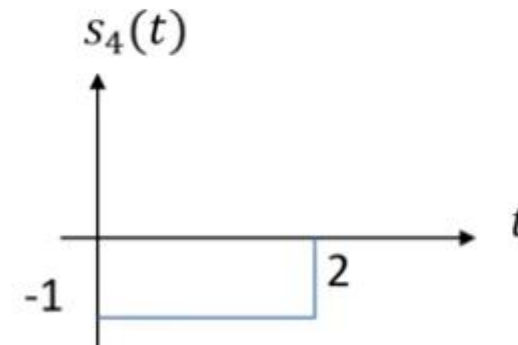
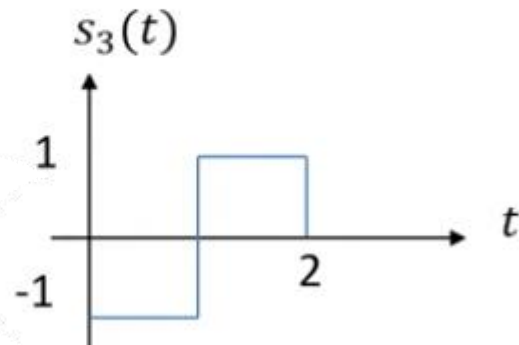
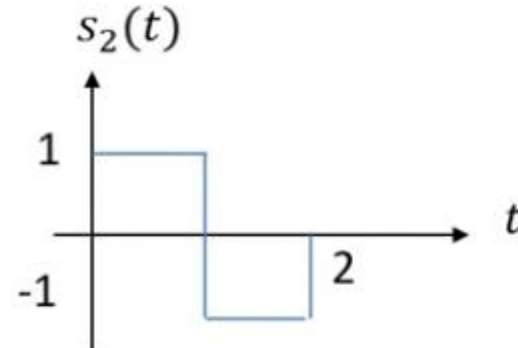
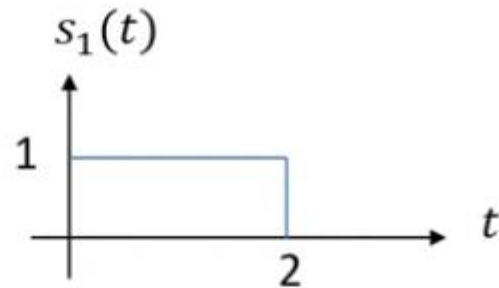
General case:

- Given a function $g_i(t)$ we can define a set of basis functions, which form an orthogonal set, as:

$$\phi_i(t) = \frac{g_i(t)}{\sqrt{\int_0^T g_i^2(t) dt}} , \quad i = 1, 2, \dots, N \quad (27)$$

Example: Gram-Schmidt Procedure

- Use Gram-Schmidt procedure to find the orthonormal basis functions for the set of signals



Example Solution:

Gram-Schmidt Procedure

- The energy for $s_1(t)$ is

$$E_1 = \int_0^2 s_1^2(t) dt = 2$$

- Hence, the first orthonormal basis function is

$$\phi_1(t) = \frac{s_1(t)}{\sqrt{E_1}} = \frac{s_1(t)}{\sqrt{2}}$$

- Projection of $s_2(t)$ over the basis function $\phi_1(t)$ is

$$s_{21} = \int_0^2 s_2(t) \phi_1(t) dt = 0$$

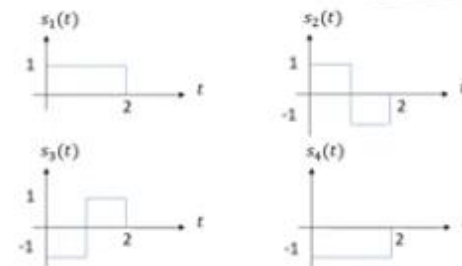
- The intermediate function

$$g_2(t) = s_2(t) - s_{21} \phi_1(t) = s_2(t)$$

$$E_{g2} = \int_0^2 s_2^2(t) dt = 2$$

- Second orthonormal basis function

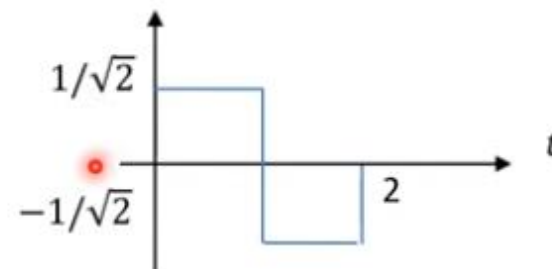
$$\phi_2(t) = \frac{g_2(t)}{\sqrt{E_{g2}}} = \frac{s_2(t)}{\sqrt{2}}$$



$\phi_1(t)$



$\phi_2(t)$



Continue Example Solution

- The intermediate function

$$s_{31} = \int_0^2 s_3(t) \phi_1(t) dt = 0$$

$$s_{32} = \int_0^2 s_3(t) \phi_2(t) dt = -\sqrt{2}$$

- Hence, $s_3(t) = s_{32} \phi_2(t)$ [No new basis function]

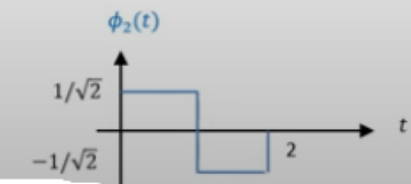
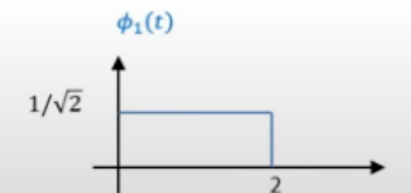
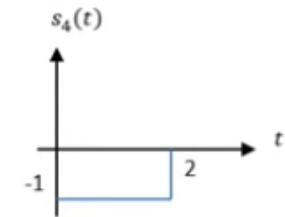
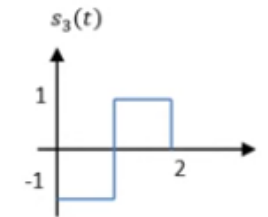
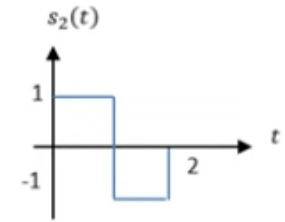
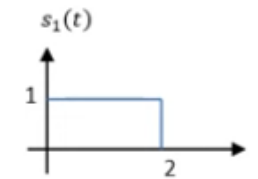
$$s_{41} = \int_0^2 s_4(t) \phi_1(t) dt = -\sqrt{2}$$

$$s_{42} = \int_0^2 s_4(t) \phi_2(t) dt = 0$$

- The intermediate function

$$g_4(t) = s_4(t) - s_{41} \phi_1(t) = 0$$

$$s_4(t) = s_{41} \phi_1(t) \text{ [No new basis function]}$$

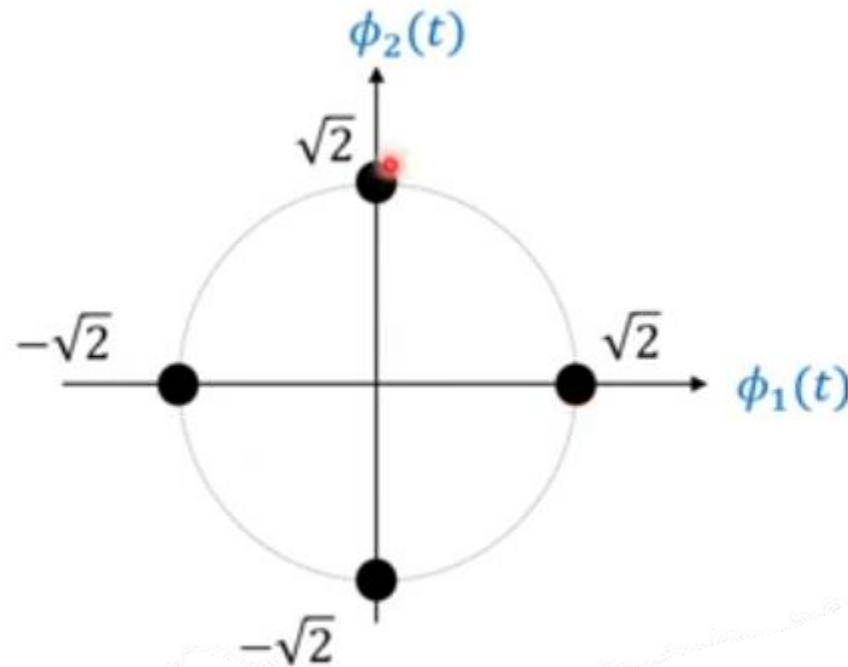


Continue Example Solution

Gram-Schmidt Procedure

- There are only two orthonormal basis function to represent the four signals $s_1(t)$, $s_2(t)$, $s_3(t)$, and $s_4(t)$ [$N = 2 < M$]

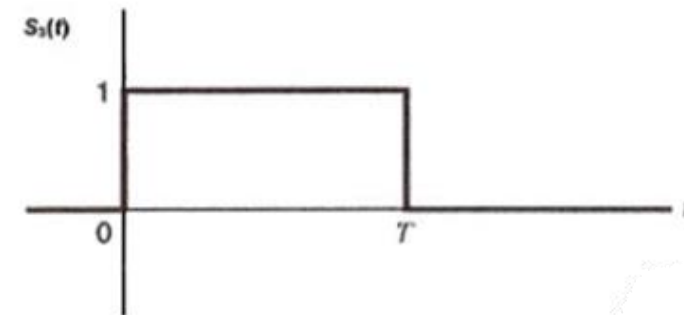
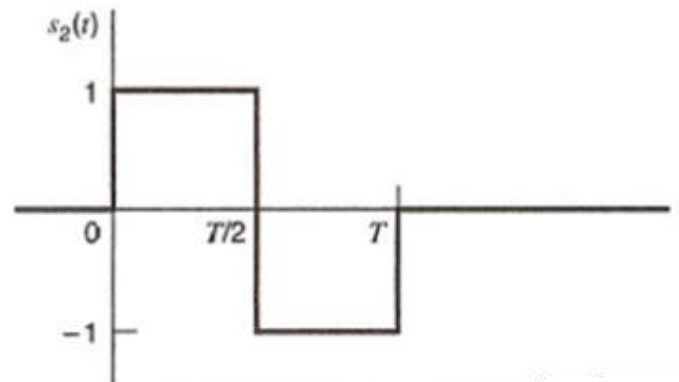
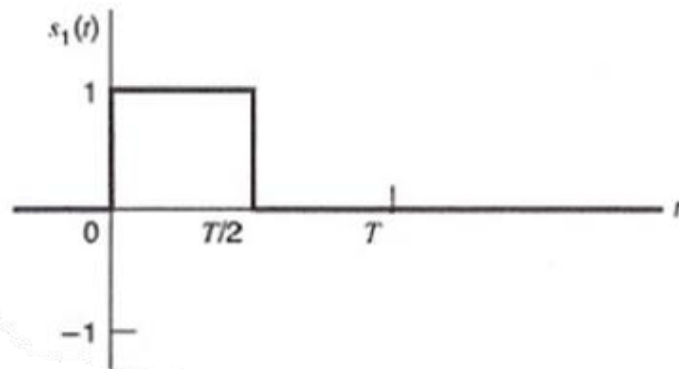
Signal Constellation Diagram (vector representation)



Practice: Gram Schmidt orthogonalization procedure

The figure below displays the waveforms of three signals.

- Using Gram Schmidt orthogonalization procedure, find an orthonormal basis for this set of signals.
- Represent the three signals in terms of the basis and show them as vectors in the signal-space diagram.



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Conversion of the Continuous AWGN Channel into a Vector Channel

- Suppose that the $s_i(t)$ is not any signal, but specifically the signal at the receiver side, defined in accordance with an AWGN channel:
- So the output of the correlator (Fig. 5.3b) can be defined as:

$$x(t) = s_i(t) + w(t),$$
$$\left\{ \begin{array}{l} 0 \leq t \leq T \\ i=1,2,\dots,M \end{array} \right\} \quad (28)$$

$$x_i = \int_0^T x(t)\phi_j(t)dt$$
$$= s_{ij} + w_i,$$
$$j=1,2,\dots,N \quad (29)$$

$$s_{ij} = \int_0^T s_i(t) \phi_i(t) dt \quad (30)$$

$$w_i = \int_0^T w(t) \phi_i(t) dt \quad (31)$$

deterministic quantity

random quantity

contributed by the
transmitted signal $s_i(t)$

sample value of the
variable W_i due to noise

Now,

- Consider a random process $\mathbf{X}^1(\mathbf{t})$, with $\mathbf{x}^1(\mathbf{t})$, a sample function which is related to the received signal $\mathbf{x}(\mathbf{t})$ as follows:
- Using 28, 29 and 30 and the expansion 5 we get:

$$x'(t) = x(t) - \sum_{j=1}^N x_j \phi_j(t) \quad (32)$$

$$x'(t) = x(t) - \sum_{j=1}^N (s_{ij} + w_j) \phi_j(t)$$

$$= w(t) - \sum_{j=1}^N w_j \phi_j(t)$$

$$= w'(t) \quad (33)$$

which means that the sample function $x^1(t)$ depends only on the channel noise!

- The received signal can be expressed as:

$$\begin{aligned}x(t) &= \sum_{j=1}^N x_j \phi_i(t) + x'(t) \\ &= \sum_{j=1}^N x_j \phi_i(t) + w'(t)\end{aligned}\tag{34}$$

NOTE: *This is an expansion similar to the one in 5 but it is random, due to the additive noise.*

Statistical Characterization

- The received signal (output of the correlator of Fig.5.3b) is a random signal. To describe it we need to use statistical methods – mean and variance.
- The assumptions are:
 - $X(t)$ denotes a random process, a sample function of which is represented by the received signal $x(t)$.
 - $X_j(t)$ denotes a random variable whose sample value is represented by the correlator output $x_j(t)$, $j = 1, 2, \dots, N$.
 - We have assumed AWGN, so the noise is Gaussian, so $X(t)$ is a Gaussian process and being a Gaussian RV, X_j is described fully by its mean value and variance.

Mean Value

- Let W_j denote a random variable, represented by its sample value w_j , produced by the j^{th} correlator in response to the Gaussian noise component $w(t)$.
- So it has zero mean (by definition of the AWGN model)

- ...then the *mean of X_j* depends only on s_{ij} :

$$\begin{aligned}\mu_{x_j} &= E[X_j] \\ &= E[s_{ij} + W_j] \\ &= s_{ij} + E[W_j] \\ \mu_{x_j} &= s_{ij} \quad (35)\end{aligned}$$

Variance

- Starting from the definition, we substitute using 29 and 31

$$w_i = \int_0^T w(t) \phi_i(t) dt \quad (31)$$

$$\begin{aligned} \sigma_{x_i}^2 &= \text{var}[X_j] \\ &= E[(X_j - s_{ij})^2] \\ &= E[W_j^2] \end{aligned} \quad (36)$$

$$\begin{aligned} \sigma_{x_i}^2 &= E \left[\int_0^T W(t) \phi_j(t) dt \int_0^T W(u) \phi_j(u) du \right] \\ &= E \left[\int_0^T \int_0^T \phi_j(t) \phi_i(u) W(t) W(u) dt du \right] \end{aligned} \quad (37)$$

$$\sigma_{x_i}^2 = \int_0^T \int_0^T \phi_i(t) \phi_j(u) E[W(t)W(u)] dt du$$

$$= E \left[\int_0^T \int_0^T \phi_j(t) \phi_i(u) R_w(t, u) dt du \right] \quad (38)$$

Autocorrelation function of the noise process

- It can be expressed as:
(because the noise is stationary and with a constant power spectral density)

$$R_w(t, u) = \frac{N_0}{2} \delta(t - u) \quad (39)$$

- After substitution for the variance we get:

$$\begin{aligned} \sigma_{x_i}^2 &= \frac{N_0}{2} \int_0^T \int_0^T \phi_i(t) \phi_j(u) \delta(t - u) dt du \\ &= \frac{N_0}{2} \int_0^T \phi_j^2(t) dt \end{aligned} \quad (40)$$

- And since $\phi_j(t)$ has unit energy for the *variance* we finally have:

$$\sigma_{x_i}^2 = \frac{N_0}{2} \quad \text{for all } j \quad (41)$$

- *Correlator outputs, denoted by X_j have variance equal to the power spectral density $N_0/2$ of the noise process $W(t)$.*

Properties (without proof)

- X_j are mutually uncorrelated
- X_j are statistically independent (follows from above because X_j are Gaussian)
- and for a memoryless channel the following equation is true:

$$f_x(x / m_i) = \prod_{j=1}^N f_{x_j}(x_j / m_i), \quad i=1,2,\dots,M \quad (44)$$

- **Define (construct)** a vector $\mathbf{X}$ of N random variables, $X_1, X_2, \dots, X_N$, whose elements are *independent Gaussian RV* with *mean values s_{ij}* , (output of the correlator, deterministic part of the signal defined by the signal transmitted) and variance equal to $N_0/2$ (output of the correlator, random part, calculated noise added by the channel).
- then the $X_1, X_2, \dots, X_N$, elements of $\mathbf{X}$ are statistically independent.
- So, we can express the *conditional probability of X* , given $s_i(t)$ (correspondingly symbol m_i) as a product of the *conditional density functions (f_x) of its individual elements f_{x_j}* .

NOTE: This is equal to finding an expression of the *probability of a received symbol* given a *specific symbol was sent*, assuming a memoryless channel

- ...that is:

$$f_x(x / m_i) = \prod_{j=1}^N f_{x_j}(x_j / m_i), \quad i=1,2,\dots,M \quad (44)$$

- where, the **vector** x and the **scalar** x_j , are sample values of the random vector X and the random variable X_j .

$$f_x(x / m_i) = \prod_{j=1}^N f_{x_j}(x_j / m_i), \quad i=1,2,\dots,M \quad (44)$$

Vector x and scalar x_j are sample values of the random vector $\mathbf{X}$ and the random variable X_j

Vector x is called
observation vector
Scalar x_j is called
observable element

- Since, each X_j is Gaussian with mean s_j and variance $N_0/2$

$$f_{x_j}(x / m_i) = (\pi N_0)^{-N/2} \exp \left[-\frac{1}{N_0} (x_j - s_{ij})^2 \right], \quad \begin{matrix} j=1,2,\dots,N \\ i=1,2,\dots,M \end{matrix} \quad (45)$$

- we can substitute in 44 to get 46:

$$f_x(x / m_i) = (\pi N_0)^{-N/2} \exp \left[-\frac{1}{N_0} \sum_{j=1}^N (x_j - s_{ij})^2 \right], \quad i=1,2,\dots,M \quad (46)$$

- If we go back to the formulation of the received signal through a AWGN channel 34

$$\begin{aligned} x(t) &= \sum_{j=1}^N x_j \phi_j(t) + x'(t) \\ &= \sum_{j=1}^N x_j \phi_j(t) + w'(t) \end{aligned} \quad (34)$$

The vector that we have constructed fully defines this part

Only projections of the noise onto the basis functions of the signal set $\{s_i(t)\}_{i=1}^M$ affect the significant statistics of the detection problem

Finally,

- The AWGN channel, is equivalent to an N-dimensional vector channel, described by the observation vector

$$x = s_i + w, \quad i = 1, 2, \dots, M \quad (48)$$

Outline

- Introduction
- Geometric Representation of Signals
 - Gram-Schmidt Orthogonalization Procedure
- Conversion of the AWGN into a Vector Channel
- *Maximum Likelihood Decoding*
- Correlation Receiver
- Probability of Error

Likelihood Functions

- In a sense, **likelihood** works *backwards* from **probability**: given parameter B , we use the conditional probability $P(A|B)$ to reason about outcome A , and given outcome A , we use the likelihood function $L(B|A)$ to reason about parameter B . This mode of reasoning is formalized in **Bayes' theorem**:

$$P(B | A) = \frac{P(A | B) P(B)}{P(A)}$$

- A **likelihood function** is a **conditional probability function** considered as a function of its *second* argument with its first argument held fixed, thus:

and also any other function proportional to such a function. That is, the likelihood function for B is the **equivalence class** of functions $b \mapsto P(A | B = b)$,

$$L(b | A) = \alpha P(A | B = b)$$

Likelihood Functions

- As we discussed in the previous class, the conditional probability density functions $f_X(\mathbf{x}|m_i)$, $i = 1, 2, 3, \dots, M$ are the very characterization of the AWGN channel.
- They express the **functional dependence** of the **observation vector** $\mathbf{x}$ on the **transmitted message symbol** m_i . (known as the transmitted message symbol)

However,

- If we have the observation vector given, and we want to define the transmitted message signal, then we have the reverse situation
- We introduce the “likelihood function” $L(m_i)$ as:

Did I drive

Looks very similar????

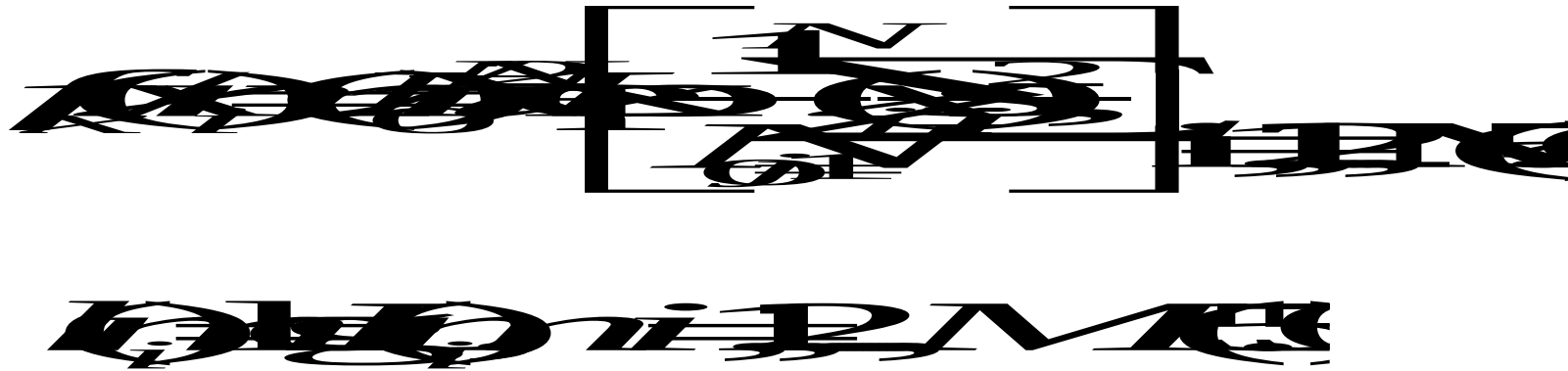
Yes, but meaning is different...

- Or log likelihood function $\ln l(m_i)$ as:

Did I drive

Log-Likelihood Function of AWGN Channel

- Substitute 46 into 5.50:



- where s_{ij} , $j = 1, 2, 3, \dots, N$ are the elements of the **signal vector** s_i , representing the **message symbol** m_i .

So,

$$-\frac{1}{2N} \sum_{n=0}^{N-1} \left(\frac{y[n] - \sum_{k=0}^{K-1} h_k x[n-k]}{\sigma^2} \right)^2$$

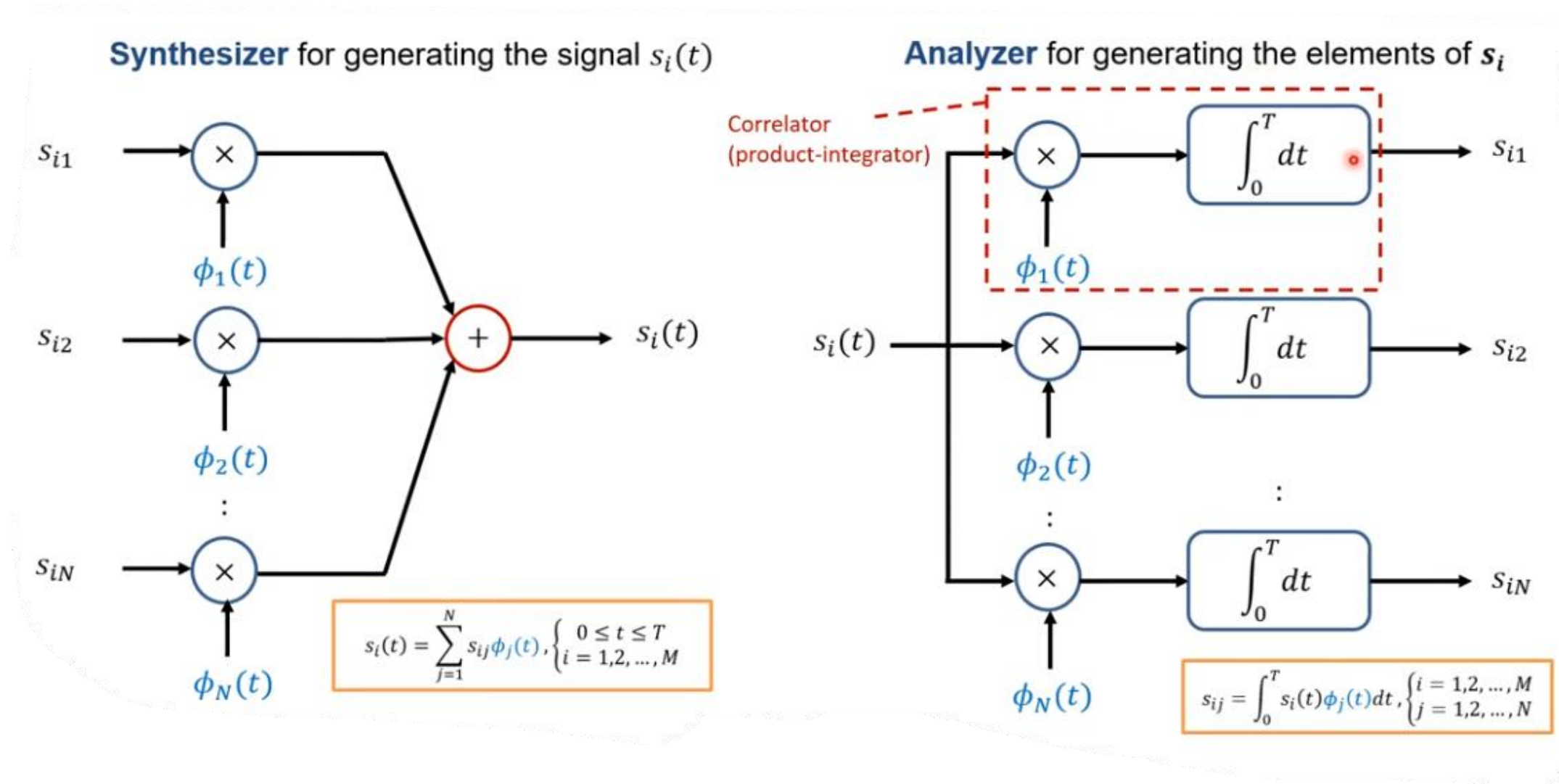
which is the log likelihood function of the AWGN channel..

Maximum Likelihood Decoding

- Defining the problem
 - Suppose that in each time slot duration of T seconds, one of M possible signals, $s_1(t)$, $s_2(t)$, ..., $s_M(t)$ is transmitted with equal probability, $1/M$.
 - As described in the previous part, for the vector representation, the signal $s_i(t)$, $i=1, 2, \dots, M$ is applied to a bank of correlators, with a common input and supplied with a suitable set of N orthogonal basis functions, N .
 - The resulting output defines the signal vector $\mathbf{s}_i$.
 - We represent each signal $s_i(t)$ as a point in the Euclidian space, $N \leq M$ (referred to as *transmitted signal point* or *message point*).
 - The set of message points corresponding to the set of transmitted signals $s_i(t)$ $\{i = 1 \text{ to } M\}$ is called a *signal constellation*.

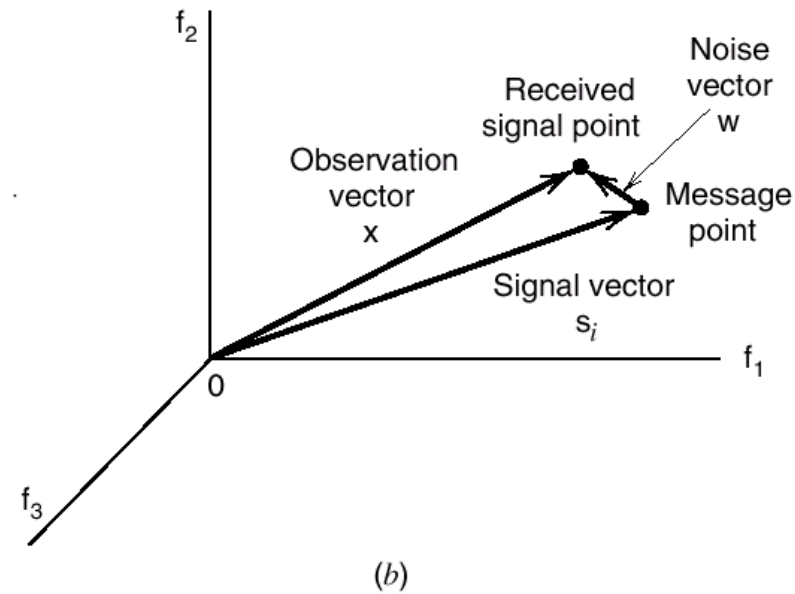
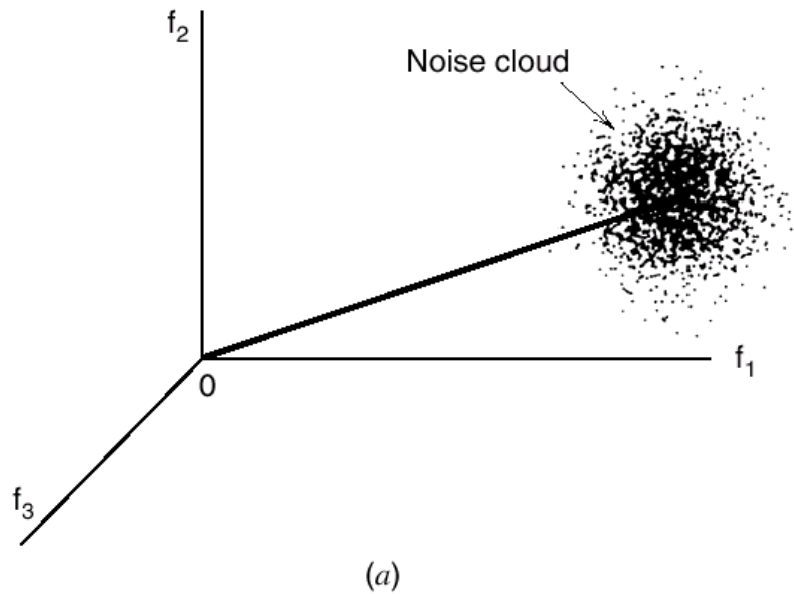
(a) Synthesizer for generating the signal $s_i(t)$.

(b) Analyzer for generating the set of signal vectors $\{s_i\}$.



- The received signal $\mathbf{x}(t)$ is applied to a bank of N correlators (Fig. 5.3b) and the correlator outputs define the *observation vector $\mathbf{x}$* .
- On the receiving side the representation of the received signal $\mathbf{x}(t)$ is complicated by the additive noise $\mathbf{w}(t)$.
- As we discussed the previous class, the *vector $\mathbf{x}$ differs from the vector $\mathbf{s}_i$* by the noise vector $\mathbf{w}$.
- However only the portion of it which interferes with the detection process is of importance to us, and this is fully described by $\mathbf{w}(t)$.

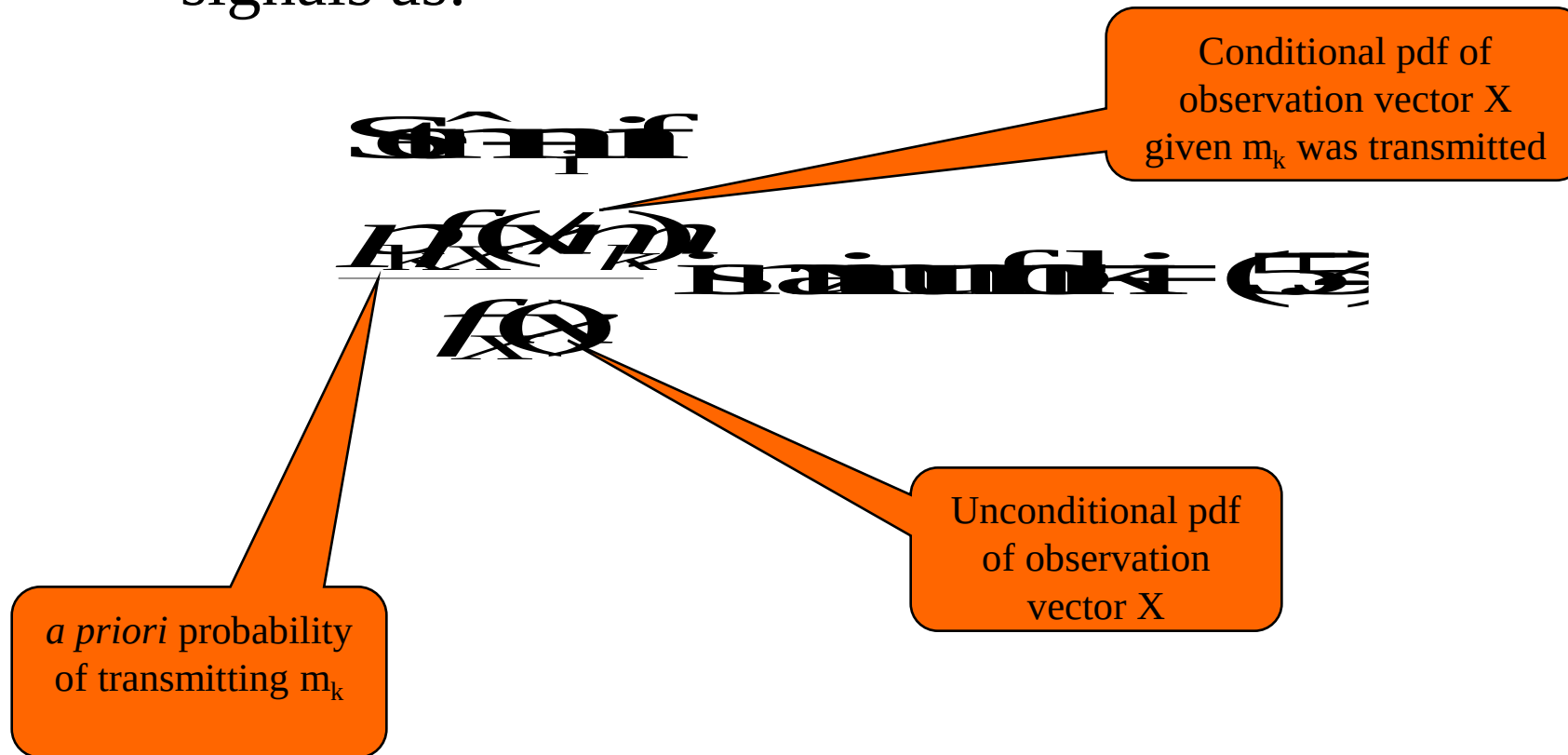
- Based on the *observation vector* $\mathbf{x}$ we may represent the received signal signal $\mathbf{x}(t)$ by a point in the same Euclidian space used to represent the transmitted signal.



- For a given observation vector x we have to make a decision $m' = m_i$
- The decision is based on the criterion to minimize the probability of error in mapping each observation vector into a decision.
- So the optimum decision rule is:

~~Step 1: Find the optimum decision rule~~

- The same rule can be more explicitly expressed using the *a priori* probabilities of the transmitted signals as:



- Thus we can conclude, according to the definition of likelihood functions, the likelihood function $l(m_k)$ will be maximum for $k = i$.
- So the decision rule using the likelihood function will be formulated as:

$$\hat{m}_i = \underset{k}{\operatorname{argmax}} l(m_k)$$

- For a graphical representation of the maximum likelihood rule we introduce the following:
 - Observation space – Z , N -dimensional, consisting of all possible observation vectors x
 - Z is partitioned into M decision regions, $Z_1, Z_2, \dots, Z_M$

$$Z = \bigcup_{k=1}^M Z_k$$

For the AWGN channel..

- Based on the log-likelihood function, of the AWGN channel, $l(m_k)$ will be max when the term: $\sum_{j=1}^N (x_j - s_{ij})^2$, is minimized by $k = i$.

- Decision rule for AWGN:

$$\sum_{j=1}^N (x_j - s_{ij})^2 \text{ is minimum} \quad (5)$$

- Or using Euclidian space notation

$$\|x - s_i\|^2 \text{ is minimum} \quad (5)$$

Finally,

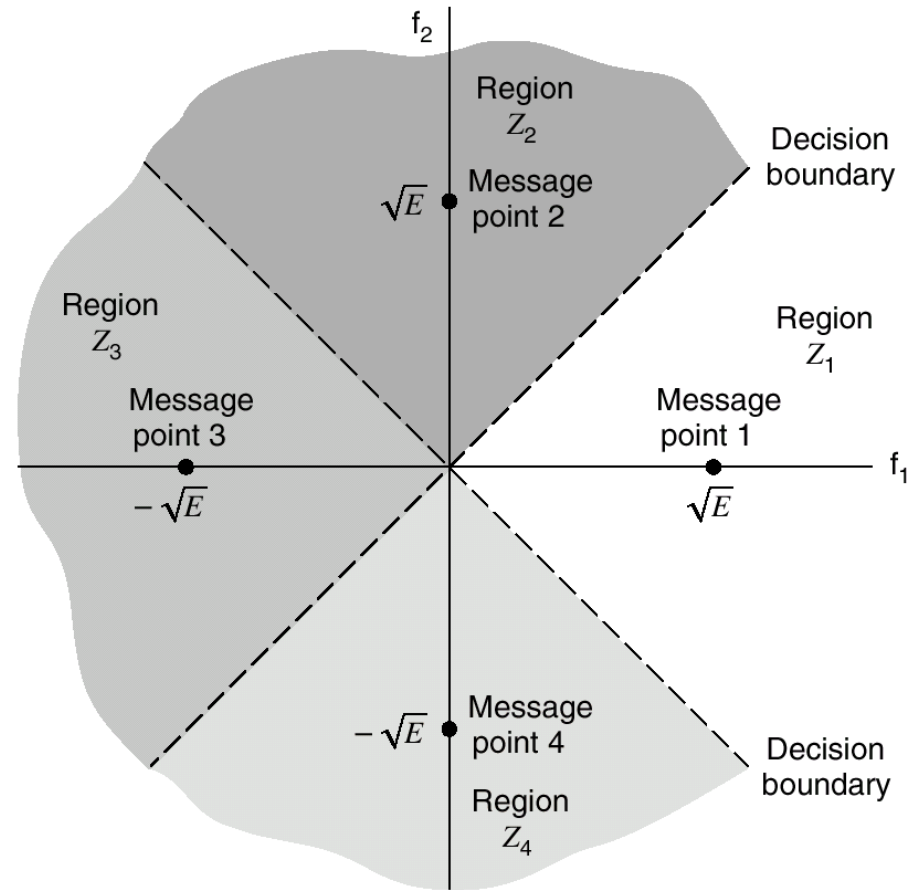
- (5.59) states that the maximum likelihood decision rule is simply to choose the message point closest to the received signal point.
- After few re-organizations we get: (*left as homework brain gymnastic exercise for you*)

~~$$\sum_{j=1}^N \frac{1}{2} s_{jk}^2$$~~ (5)

$$E_k = \sum_{j=1}^N s_{jk}^2 \quad (5)$$

Energy of the
transmitted signal
 $s_k(t)$

Illustrating the partitioning of the observation space into decision regions for the case when $N = 2$ and $M = 4$; it is assumed that the M transmitted symbols are equally likely.

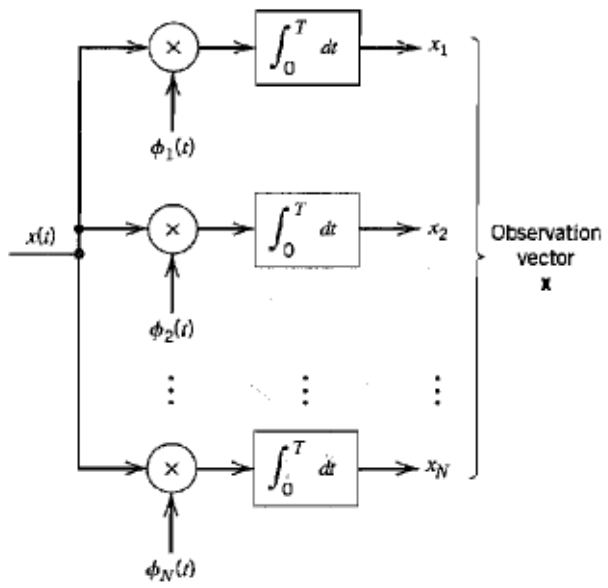


Outline

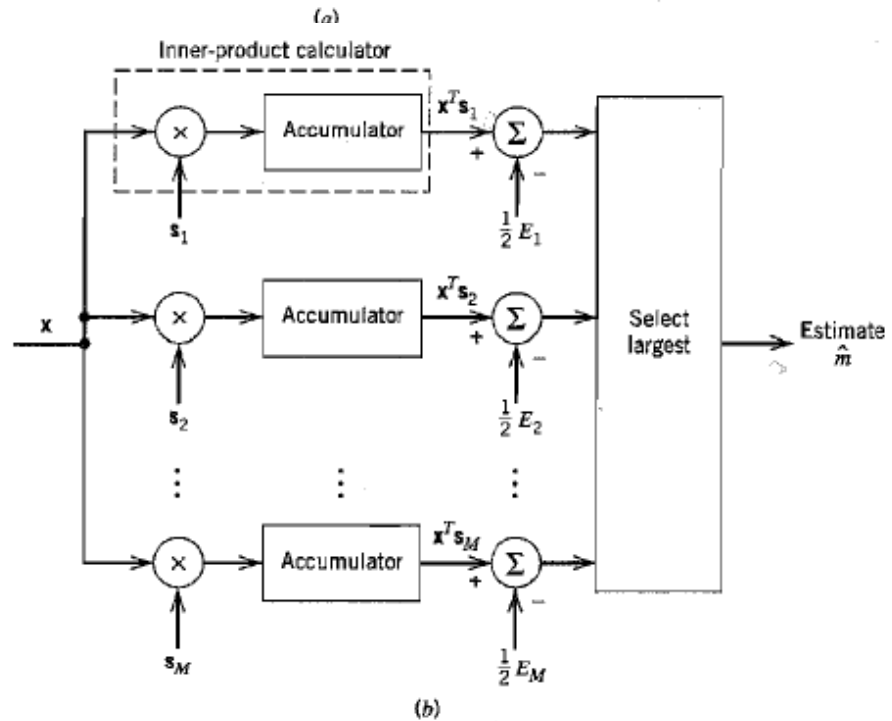
- Introduction
- Geometric Representation of Signals
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- Likelihood Functions
- Maximum Likelihood Decoding
- *Correlation Receiver*
- Probability of Error

Correlation Receiver

- Based on the theoretical assumptions made in the previous class we define the correlator at the receiver side.
- It can be implemented as a optimum receiver that consists of two parts:
 - **Detector part** – M product-integrators supplied with the corresponding set of coherent reference signals (orthogonal basis functions), generated locally. It operates on the received signal $\mathbf{s}(t)$ to produce the observation vector $\mathbf{x}$ for $0 \leq t \leq T$.
 - **Receiver part** – signal transmission decoder – which is implemented in the form of a maximum likelihood decoder, operating on the observation vector $\mathbf{x}$ to produce the estimate m' of the transmitted symbol m_i in a way to minimize the average probability of symbol error. *According to (5.61) the N elements of the observation vector $\mathbf{x}$ are multiplied by the N elements of each of the M signal vectors $s_1, s_2, \dots, s_M$ and then summed up to produce the inner products $[x^T s_k | k=1, 2, \dots, M]$. Largest of the resulting numbers is selected.*



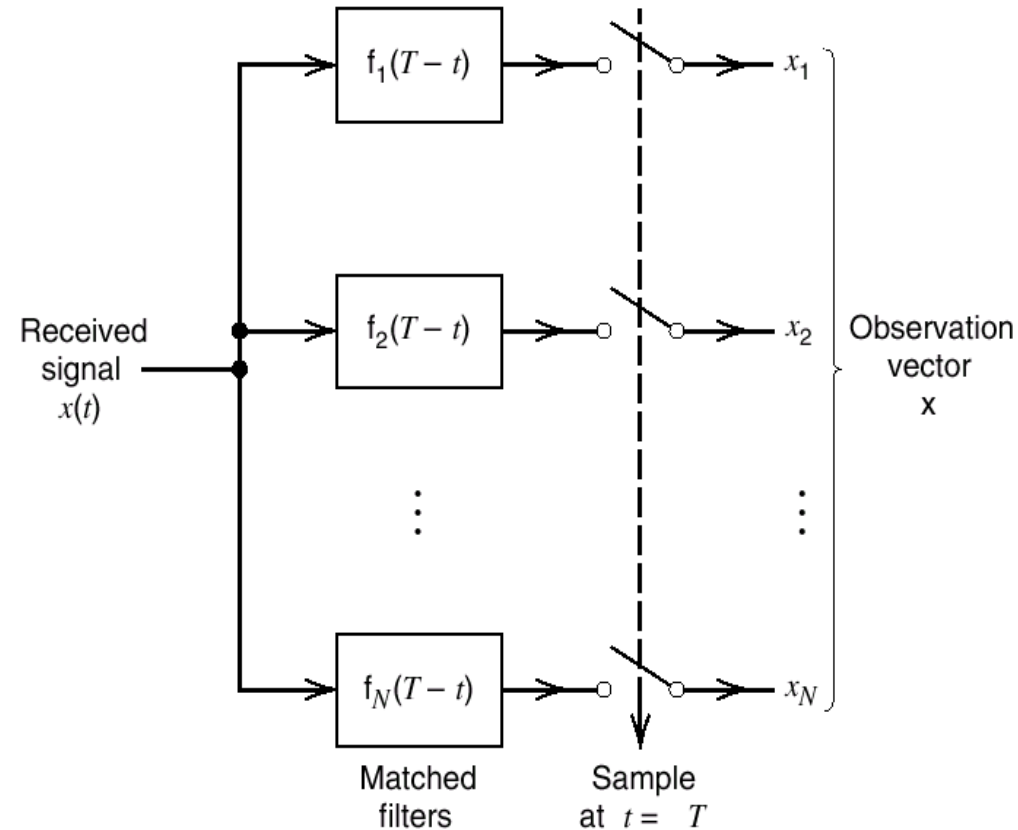
(a) *Detector or demodulator.*



(b) *Signal transmission decoder.*

Note:

- The detector shown in Fig. 5.9a is based on *correlators*.
- Alternatively, *matched filters*, may be used to produce the required *observation vector $\mathbf{x}$* .



Detector part of matched filter receiver;
the signal transmission decoder is as shown in Fig. 5.9b

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Probability of Error

- To complete the statistical characterization of the correlation receiver (Fig. 5.9) we need to discuss its *noise performance*.
- Using the assumptions made before, we can define the average probability of error P_e as:

$$\begin{aligned}
 P_e &= \sum_{j=1}^M p_j \text{Pr}(\text{error} | \mathbf{z}_j | \mathbf{g}, \mathbf{a}, \mathbf{n}) \\
 &= \frac{1}{M} \sum_{j=1}^M \text{Pr}(\text{error} | \mathbf{z}_j | \mathbf{g}, \mathbf{a}, \mathbf{n}) \quad (56) \\
 &= 1 - \frac{1}{M} \sum_{j=1}^M \text{Pr}(\text{decision} | \mathbf{z}_j | \mathbf{g}, \mathbf{a}, \mathbf{n})
 \end{aligned}$$

- Using the likelihood function this can be re-written as:

$$P_e = \frac{1}{M} \sum_{m=1}^M \sum_{n \neq m}^M \frac{d_{mn}}{d_{m, \text{min}}}$$

- The probability of error is invariant to rotation and translation of the signal constellation.
 - In maximum likelihood detection the probability of symbol error P_e depends solely on the Euclidian distances between the message points in the constellation
 - The additive Gaussian noise is spherically symmetric in all directions in the signal space.

Summary

- This lecture presents a systematic procedure for the analysis of *signals in a vector space*.
- The basic idea of the approach is to represent each member of a set of transmitted signals by an N-dimensional vector, where N is the number of orthogonal basis function, needed for the unique representation of the transmitted signals.
- The set of signal vectors defines the *signal constellation*, the N-dimensional space defines the *signal space*.
- It is the theoretical basis for the design of a digital communication receiver in the presence of AWGN. The procedure is based on the theory of maximum likelihood detection.
- The *average probability of symbol error* is defined as P_e . It is dominated by the nearest neighbors to the transmitted signal.

Thank You